

USER MANUAL

Command Dashboard & Diagnostic System

Welcome to the Coach Maintenance System powered by AI and Computer Vision. This tool is designed for railway engineers to perform high-speed, accurate diagnostics of bogie components.

1. System Overview

This system evaluates uploaded images of train bogie components to classify them as:

- FIT (Normal / Safe for operation)
- UNFIT (Defect / Requires replacement)
- MONITOR (Borderline / Keep under observation)

The AI models have been specifically trained for 6 critical components:

1. CBC Coupler
2. Axle Box
3. Brake Disk
4. Damper
5. Coil Spring
6. Wheel

2. Navigating the Interface

The application interface is divided into two main areas:

Sidebar (Left Panel)

Contains the main navigation tabs:

- Dashboard: Overview of total inspections and AI training status.
- Component Inspector: The main workspace for running diagnostics.
- Reports: Access to generated inspection history.

The sidebar can be collapsed using the menu button at the top left.

Topbar (Header)

- Features the current page title and a live system clock.
- On the right, you will see the Counterpoint partner logo.
- AI Status Badge: A pill-shaped badge showing the number of ready AI models (e.g., 'AI 6/6'). Hover over this badge to see the detailed training status of each component. A green dot indicates all models are loaded and ready.

3. Running a Diagnostic (Component Inspector)

The core function of the software is the Component Inspector. Follow these steps to inspect a part:

Step 1: Select the Component

Use the tabs or dropdown menu at the top of the Inspector page to select the component you are evaluating (e.g., 'Axle Box' or 'Brake Disk').

Step 2: Upload an Image

Click or drag-and-drop an image into the upload area.

Note: Ensure the image is well-lit and clearly focuses on the component for the best AI accuracy.

Step 3: Run Diagnostic

Click the 'Run Diagnostic' button. The system will process the image through two layers:

1. Traditional Computer Vision (CV): Extracts edge features, rust thresholds, and structural alignment.
2. MobileNetV2 Deep Learning (AI): Analyzes the patterns to make a final prediction.

Step 4: Interpret the Results

Once processing is complete, the dashboard will display:

- The Original Image.
- A Heatmap Overlay highlighting the areas the AI focused on.
- The Final Classification (FIT, UNFIT, or MONITOR).
- AI Confidence Score (percentage of certainty).

4. Generating & Exporting Reports

After running a diagnostic, you can export the results as a shareable document.

1. Locate the 'Export / Share Report' section at the bottom of the Inspector page.
2. Add any manual mechanic notes or observations in the provided text field.
3. Click 'Download PDF Report'.
4. The system will bundle the images, AI scores, and your notes into a professionally formatted PDF branded with the logo. This file is automatically saved to your computer and can be shared via email or printed for records.

5. Troubleshooting & FAQ

Issue: The AI Status Badge says 'NOT TRAINED' for a component.

Solution: This means the backend server has not loaded the `.pth` weight file for that model. Ensure the backend server is running and the models exist in the `backend/` directory.

Issue: The Heatmap is completely blank.

Solution: The uploaded image might be too dark or out of focus. Try uploading a clearer image. The AI relies on distinct edge and color data to generate the Grad-CAM heatmap.

Issue: Cannot connect to the server (Network Error).

Solution: Verify that the FastAPI backend server is running in your terminal (usually on port 8001). Check your network connection.

6. Support & Contact

For technical assistance, system upgrades, or bug reports, please contact the IT administration desk.

System Developed for:

Indian Railways

RDSO Compliant

Powered by Counterpoint Technology.